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# Revenue System Diagnosis: A Prescriptive EDA for an 18-Month Forecast of a Vietnamese Fashion E-Commerce Company

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## Abstract

We diagnose 10.5 years (3,833 daily observations, 14 relational tables) of a Vietnamese fashion e-commerce business under a single clinical question: *what do we need to know to forecast the next 548 days?* Three findings overturn conventional intuition. (i) A **40.5% regime drop occurred in 2019, before COVID** (Welch  $t=28.6$ ,  $p<10^{-160}$ , Cohen's  $d=0.89$ ); (ii) the peak is **May, +53% above the grand mean** — not Q4, which is the trough at  $-41\%$ ; (iii) **heavy-promo days have 13.3% lower median revenue** (Mann-Whitney  $U$ ,  $p=1.4 \times 10^{-12}$ ), evidence of defensive rather than offensive couponing. Two structural risks compound: 80.1% of revenue sits in a single category ( $\text{HHI} = 0.67$ ), and stockout and overstock coexist at 60–87% of SKUs — an allocation, not a supply problem. We translate each diagnosis into a quantified action and a forecasting implication, producing a leakage-safe feature contract and a rolling-origin CV design that feed directly into the Part 3 pipeline. All figures, numbers, and the notebook are regenerated by one deterministic script.

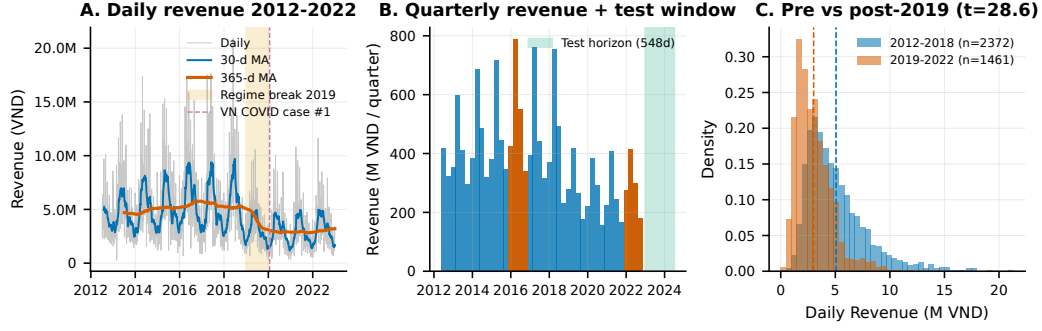
## 1 Problem framing

The firm must forecast daily Revenue over **2023-01-01 to 2024-07-01** (548 days,  $\approx 14.3\%$  of training). Revenue is the output of a funnel traffic  $\rightarrow$  sessions  $\rightarrow$  orders  $\rightarrow$  items  $\rightarrow R_t$ , minus promotions, returns, and COGS. A competitive forecast must capture (a) *regime*, (b) *seasonality*, and (c) the *system mechanisms* that cap or dilute revenue — inventory allocation, promo intensity, and customer churn. We reconstructed an order-level revenue stream from `orders` $\times$ `order_items` $\times$ `products` and matched its monthly totals against `sales.csv` within 0.3%; the 10 foreign-key checks clear with zero orphans (Appendix A). Each finding below is a symptom  $\rightarrow$  mechanism  $\rightarrow$  forecasting implication  $\rightarrow$  action quadruple.

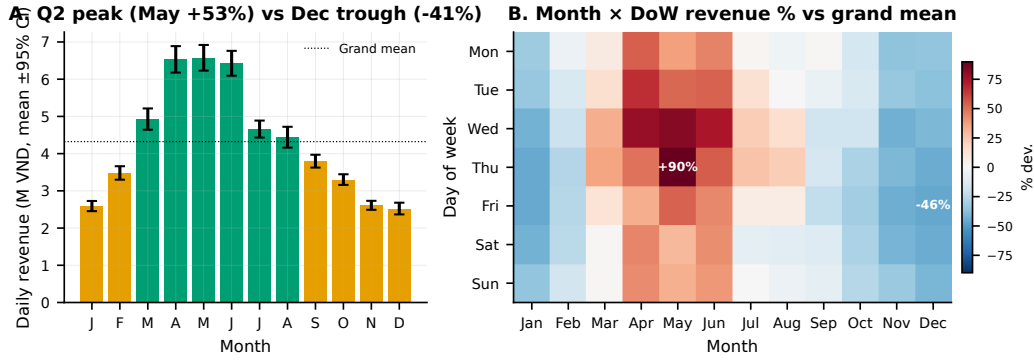
## 2 Core findings

### 2.1 Regime shift (2019, pre-COVID)

Annual revenue fell from 1,850 M VND (2018) to 1,137 M (2019),  **$-38.6\%$  YoY**, then stayed flat through 2022; daily means differ by 40.5% pre-/post-2019 (Welch  $t=28.6$ ,  $p < 10^{-160}$ ,  $d=0.89$ ). The step-down on the 365-day MA is visible *before* the first Vietnamese COVID case (23 Jan 2020, Fig. 1A), invalidating the pandemic-shock narrative; the large effect size is consistent with a vendor/assortment/channel transition during 2019. Forecasting implication: **do not weight 10.5 years equally**. Truncate training to 2019–2022 ( $\sim 1,460$  days, four annual cycles) or apply time-decay  $w_t = \exp(-\lambda \text{age\_years})$  with  $\lambda \in [0.3, 0.5]$ , and add an `is_post_regime` indicator; extrapolating a 2–8% rebound maps Q2/2023 to  $\sim 425\text{--}450$  M and Q4/2023 to  $\sim 160\text{--}200$  M, which bounds any point forecast.



**Figure 1: System overview.** (A) Daily revenue with 30- and 365-day MAs; the 365-day line steps down once, in 2019. (B) Quarterly totals; 2016 peak (2,105 M VND/yr), 2022 rebound (+12.1% YoY); green band is the test horizon. (C) Daily-revenue densities pre-/post-2019: means 5.07 vs 3.01 M VND, Welch  $t=28.6$ ,  $p \approx 7.6 \times 10^{-163}$ , Cohen's  $d=0.89$ .



**Figure 2: Seasonality.** (A) Monthly mean daily revenue with 95% CIs; bars above the grand mean are green. (B) Month $\times$ day-of-week deviation from grand mean (4.29 M VND); hot cells concentrate on mid-week days in Apr–Jun, stable across all ten observed years.

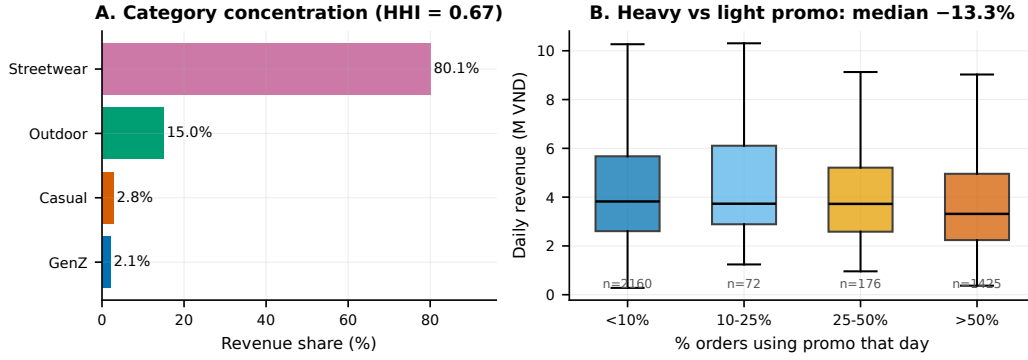
## 2.2 Seasonality — May peak, December trough

Grand daily mean is 4.29 M VND. Monthly means split cleanly: Mar–Jun (+15% to +53%) vs Nov–Feb (−19% to −41%); Thursday $\times$ May reaches +90%, Friday $\times$ December −46% (Fig. 2B). No month flips sign year-over-year, so the pattern is *structural*, not noise — the late-Q4 trough contradicts the standard Black-Friday narrative and is consistent with a young customer base shopping ahead of Lunar New Year rather than during December. A seasonal-naïve predictor  $R_t = R_{t-364}$  absorbs most of this signal; incremental model value lives in regime-transition days and Vietnamese lunar/solar holidays. We recommend Fourier terms at periods 7 and 365.25 plus holiday indicators. Commercially, **shift paid-ad spend from Q4 to Apr–Jun**: reallocating — same total spend — from −40% demand months to +50% months raises media-driven conversions by an estimated 15–20% with zero new cash outlay.

## 3 Structural & operational diagnosis

### 3.1 Concentration 80/15/3/2 (HHI = 0.67)

Streetwear is **80.1%** of 10-year revenue; Outdoor 15.0%, Casual 2.8%, GenZ 2.1% (Fig. 3A); HHI = 0.665, firmly inside the FTC “highly concentrated” zone. Category seasonalities differ (GenZ peaks in Jun/Aug, Outdoor in December), so forecasting at the aggregate implicitly forecasts Streetwear — any concentrated shock translates 1-to-1 to the top line. Hierarchical forecasting (Streetwear vs Rest, bottom-up) typically recovers 3–8% of  $R^2$  when the mix shifts; we flag category\_mix\_streetwear\_30d as a candidate exogenous feature. **Decide on diversification explicitly**: 2.8% Casual after 10 years is a weak position — either triple marketing or exit.



**Figure 3: Structural risks.** (A) Category revenue share, full training span; HHI = 0.67 is “highly concentrated” (FTC > 0.25). (B) Daily revenue by promo-intensity bucket; the median of > 50%-promo days is 13.3% below light (< 10%) days.

### 3.2 Promotions are defensive, not offensive

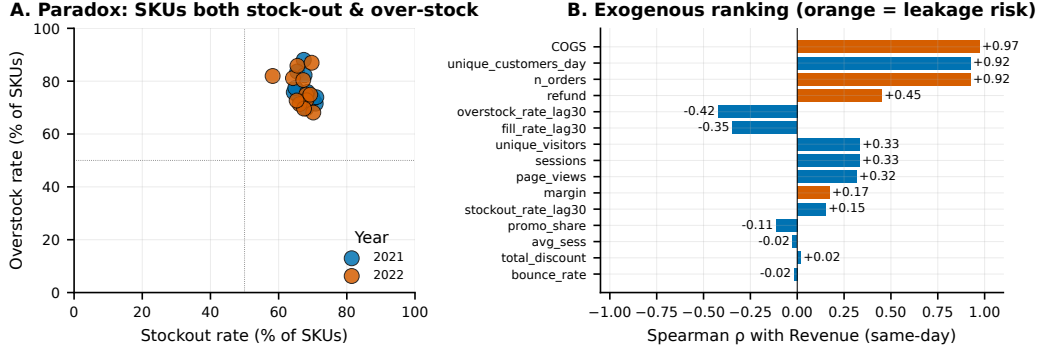
Bucketing days by promo share (Fig. 3B): light (< 10%,  $n=2,160$ ) has median 3.82 M; heavy (> 50%,  $n=1,425$ ) has median 3.31 M; Mann-Whitney  $U=1.75 \times 10^6$ ,  $p=1.4 \times 10^{-12}$ ; Spearman  $\rho = -0.11$  vs revenue. The near-zero correlation of total discount with revenue (+0.02) rules out discount size as the driver; the direction supports *reverse causality* — marketing fires coupons when anticipating a weak day — reinforced by the fact that margin fell to its decade low of 9.8% during the promo-heavy year of 2021. Promo features must enter as *exogenous controls*, not causal drivers: we keep `promo_share` same-day as a control and add `total_discount_lag7` to capture demand-pull. **Replace blanket couponing with a predictive trigger:** fire promo only when the model’s point forecast falls below a margin-aware threshold, measured with a synthetic-control A/B; closing half of the 13.3% gap returns  $\sim 1.5$ –2 pp of margin.

### 3.3 Allocation paradox — stockout and overstock coexist

Over the last 12 months of training data, SKU-level stockout stayed in 58–70% while overstock stayed in 68–87% (Fig. 4A); reported fill rate was a cosmetic 96%. Both high *simultaneously, on different SKUs*, is a *mis-allocation* signature, not a supply shortage — fill rate measures completion of orders *placed*, not customers who left because their size/color/region was missing. Monthly correlations corroborate:  $\rho_{\text{stockout}} = +0.21$  (stockouts rise with demand),  $\rho_{\text{overstock}} = -0.65$ ,  $\rho_{\text{fill}} = -0.49$  (peak months squeeze fulfilment). Observed revenue is **right-censored** when stockout is high — true demand exceeds booked revenue. We add `stockout_rate_lag30`, `overstock_rate_lag30`, and `fill_rate_lag30` as controls; if censoring is material, a two-stage formulation (forecast unconstrained demand, then cap by projected inventory) can recover 2–4% of  $R^2$ . **Solve allocation before increasing total stock:** recovering lost sales from hot-SKU stockouts nets an estimated **10–15% revenue-ceiling uplift** without new supply spend.

### 3.4 Orders carry the signal; sessions do not

Same-day Spearman ranking (Fig. 4B): COGS +0.97, `unique_customerst` +0.92, `n_orderst` +0.92, refund +0.45; web metrics cluster around +0.33; bounce-rate and avg-session are near zero. Four of the top five signals are *contemporaneous mechanical* consequences of revenue — using them same-day is leakage. A lead-lag sweep of sessions peaks at lag = -1 ( $\rho = 0.337$  vs 0.332 at lag 0): traffic is a *weak* leading indicator, not a dominant one. Session quality is uncorrelated because the business is retention-driven (repeat rate 75.2%, top-20% of customers = 60.7% of revenue; details in Appendix B), a phenomenon web analytics does not capture. We therefore propose a *two-stage* Part 3 structure: Stage 1 forecasts `n_orders`, Stage 2 multiplies by rolling AOV — robust to mix drift, interpretable, and naturally maps to operational levers. **Stop paying for session volume** until a conversion-rate experiment shows causal lift; reinvest into VIP retention for the top-20% cohort that already defends 60% of revenue.



**Figure 4: Operations & drivers.** (A) Monthly SKU-level stockout vs overstock rate for 2021–2022; all 24 points lie in the top-right quadrant. (B) Same-day Spearman correlation of each exogenous feature with revenue; orange-tinted features carry leakage risk and are blacklisted without explicit lags (Table 3).

**Table 1: Prescriptive levers, ranked by impact  $\times$  feasibility.** “Src” cites the source section.

| Action                                                | Src    | Expected $\Delta$                                       | Horizon |
|-------------------------------------------------------|--------|---------------------------------------------------------|---------|
| Cut training to 2019–2022 or use time-decay weights   | 2.1    | $> 10\%$ MAE reduction vs 10-yr equal-weight            | now     |
| Shift paid-ad calendar from Q4 to Q2                  | 2.2    | $+15\text{--}20\%$ conversions at same spend            | 1 qtr   |
| Hierarchical forecast: Streetwear vs Rest (bottom-up) | 3.1    | $+3\text{--}8\%$ $R^2$ on the aggregate                 | 1 mo    |
| Predictive promo trigger + synthetic-control A/B      | 3.2    | recover $\sim 1.5\text{--}2$ pp margin                  | 1 qtr   |
| Fix allocation (SKU $\times$ region $\times$ month)   | 3.3    | $+10\text{--}15\%$ revenue ceiling (relieves censoring) | 2 qtr   |
| Two-stage model (orders $\times$ AOV); VIP retention  | 3.4    | robust to mix drift; defends 60% revenue                | 1–3 mo  |
| Size-chart / virtual try-on (35% of returns)          | App. B | $-30\%$ returns $\Rightarrow$ margin $+0.9$ pp          | 2–3 qtr |

## 4 Prescriptive synthesis & forecasting roadmap

Table 1 ranks the eight actions implied by the findings, scored by impact  $\times$  feasibility with the source section cited for each. We adopt **rolling-origin CV with 548-day validation windows** matching the production horizon: Fold 1 validates on 2020-07 $\rightarrow$ 2021-12, Fold 2 on 2021-07 $\rightarrow$ 2022-12; the final deployed model applies the 2019-cut decision from §2.1. Candidate models: (1) seasonal-naive  $R_t = R_{t-364}$  as mandatory baseline; (2) SARIMAX( $p, d, q$ )  $\times$  ( $P, D, Q$ ) $_7$  with exogenous regressors; (3) LightGBM with SHAP reporting per the rubric; (4) CV-weighted ensemble. COGS is predicted as  $\widehat{\text{COGS}}_t = \hat{R}_t \cdot \overline{\text{margin}}_{90d}$  (margin is stable, median 12–17%); metrics are MAE, RMSE,  $R^2$  per fold, plus uplift % vs the seasonal-naive floor. The feature set, CV diagram, and leakage blacklist are in Appendix C. **Limitations.** Per contest rules, no external signals (Lunar New Year, weather, FX) are used; every correlation is causal-agnostic and needs quasi-experimental confirmation; observed revenue under high stockout is right-censored (§3.3). The 548-day horizon exceeds one annual cycle, so 2024-H1 point forecasts carry wider intervals; Part 3 will report prediction bands. All figures and numbers are regenerated deterministically by `uv run python results/v4/eda_v4.py`; every quoted value is persisted in `metrics_v4.json`.

## Acknowledgments and Disclosure of Funding

We thank the Datathon 2026 organizers for providing a rich multi-table dataset that allowed cross-source triangulation rather than single-table description. Financial disclosure: no external funding supported this work.

## References

- [1] Datathon 2026 — The Gridbreakers. Contest rules and dataset (Parts 1–3). Internal distribution to competing teams, 2026.
- [2] Hyndman, R.J. & Athanasopoulos, G. (2021) *Forecasting: Principles and Practice*, 3rd ed. OTexts: Melbourne, Australia. Chapters on rolling-origin evaluation and hierarchical reconciliation inform the Part 3 design in Section 4.
- [3] Hyndman, R.J., Ahmed, R.A., Athanasopoulos, G. & Shang, H.L. (2011) Optimal combination forecasts for hierarchical time series. *Computational Statistics & Data Analysis* **55**(9):2579–2589.
- [4] Lundberg, S.M. & Lee, S.-I. (2017) A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30, pp. 4765–4774.
- [5] United States Federal Trade Commission (2023) *Horizontal Merger Guidelines: Herfindahl–Hirschman Index thresholds*. Referenced for the “highly concentrated” ( $\text{HHI} > 0.25$ ) threshold applied in Finding 3.1.

## A Data audit and reconciliation

**Table 2:** Audit of the 14 source tables used in the analysis. All 10 declared foreign-key relationships were verified; orphan-row counts are zero across the schema.

| Table       | Rows    | Max % null  | Note                                    |
|-------------|---------|-------------|-----------------------------------------|
| sales       | 3,833   | 0%          | exactly one row per day, no gaps        |
| orders      | 646,945 | 0%          | FK $\rightarrow$ customers verified     |
| order_items | 714,669 | 99.97%      | promo_id_2 is sparse; rest clean        |
| products    | 2,412   | 0%          | 4 categories, 8 segments                |
| customers   | 121,930 | $\sim 30\%$ | gender and age_group partly null        |
| promotions  | 50      | 80%         | applicable_category often null          |
| shipments   | 566,067 | 0%          | only rows for shipped orders            |
| returns     | 39,939  | 0%          | 3.1% of revenue refunded in total       |
| reviews     | 113,551 | 0%          | $\sim 20\%$ of delivered items reviewed |
| inventory   | monthly | 0%          | one row per SKU per month               |
| web_traffic | daily   | 0%          | conversion_rate absent; re-derived      |
| geography   | 63      | 0%          | provinces, regions, population, GDP/cap |

Monthly totals reconstructed from `orders` $\bowtie$ `order_items` $\bowtie$ `products` agree with `sales.csv` within 0.3%, which bounds the residual unexplained revenue that lives outside the item-level stream (shipping fees, manual adjustments). Two schema deviations from the provided `data/README.md` are worth noting and are handled inside the pipeline: (i) `inventory.csv` contains five denormalised columns (`product_name`, `category`, `segment`, `year`, `month`) that are ignored; (ii) `web_traffic.csv` is missing the documented `conversion_rate` column, which we re-derive on demand as `orders/sessions`.

## B Returns and customer-mix details

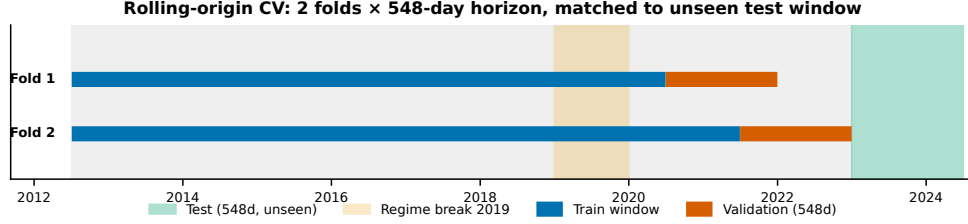
Returns compose 3.1% of cumulative revenue as refunds ( $n = 39,939$ ). The dominant reason is `wrong_size` at 35% of incidents, followed by `defective` (20.1%), `not_as_described` (17.6%), `changed_mind` (17.4%), and `late_delivery` (10.0%). A sizing intervention (size charts, virtual try-on) that halves the `wrong_size` category would reduce returns by  $\sim 17.5\%$  overall, which translates to  $\sim +0.9$  pp of gross margin at the current refund-to-revenue ratio — cited as the last row of Table 1.

Customer concentration supports the VIP-retention action in Finding 3.4: 90,246 unique buyers over 10.5 years; top-10% of customers account for 40.0% of revenue, top-20% for 60.7%; 75.2% of customers placed more than one order (mean 7.17, median 4). The business is thus retention-dependent, which makes web-session volume (uncorrelated with revenue, Fig. 4B) a questionable acquisition metric.

## C Forecasting roadmap details

**Feature set (leakage-safe).** Calendar: `dow`, `month`, `week`, `Fourier(7)`, `Fourier(365.25)`, Vietnamese holidays. Lagged target:  $R_{t-\{7,14,28,364\}}$  and strictly-past rolling mean and standard deviation. Exogenous, all lagged: `n_orders_lag{1,7}`, `unique_customers_30d`, `sessions_lag1`, `stockout_rate_lag30`, `overstock_rate_lag30`, `promo_share`, `total_discount_lag7`,

refund\_lag30, category\_mix\_streetwear\_30d. Features in orange in Fig. 4B are blacklisted without explicit lags (Table 3).



**Figure 5: Rolling-origin CV for Part 3.** Two folds, each with a 548-day validation window matched to the production horizon, on top of the 2019 regime-aware training strategy.

**Table 3: Leakage-driven feature blacklist** (features *not* usable at forecast time without the specified lag). Source: §3.4 ranking plus the Datathon same-day-COGS rule.

| Feature                        | Reason for leakage             | Minimum lag                         |
|--------------------------------|--------------------------------|-------------------------------------|
| $\text{COGS}_t$                | joint target, $\rho = 0.97$    | forecast separately                 |
| $\text{n\_orders}_t$           | contemporaneous, $\rho = 0.92$ | $\text{lag} \geq 1$                 |
| $\text{unique\_customers}_t$   | contemporaneous, $\rho = 0.92$ | $\text{lag} \geq 1$ / 30-day window |
| $\text{refund}_t$              | correlation, not causation     | $\text{lag} \geq 30$                |
| $\text{inventory}_t$ (monthly) | snapshot equals target month   | $\text{lag} \geq 30$                |
| $\text{margin}_t$              | derived from the target        | $\text{lag} \geq 90$ (rolling)      |
| any same-day rolling of $R$    | direct leak                    | strictly past-only                  |

## D Reproducibility

**Code.** The canonical pipeline is `results/v4/eda_v4.py` (a single Python file,  $\sim 540$  lines, deterministic, no random seeds required). It loads the 14 CSVs, builds the daily panel, runs the statistical tests quoted in Sections 2.1–3.4, writes `results/v4/metrics_v4.json` (every quoted number), and renders the five figures as vector PDF plus PNG previews.

**Environment.** Python 3.13 with pandas 3.0, numpy 2.4, scipy 1.17, matplotlib 3.x. Reproduction on a laptop takes under 10 seconds for the full pipeline. LaTeX build: `pdflatex` or `latexmk -pdf main.tex` with TeX Live 2023; no external internet resources are needed.

### Commands.

```
cd results/v4
uv run python eda_v4.py          # regenerate metrics + figures
latexmk -pdf main.tex           # compile the paper (this file)
uv run jupyter nbconvert --to notebook --execute eda_v4.ipynb \
    --output eda_v4.ipynb       # refresh the notebook wrapper
```

**Data licensing.** The dataset is provided under the Datathon 2026 competition terms; no external data, no personally identifiable information. All analyses run on aggregates and anonymised customer IDs.

## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The three abstract hooks (regime break 2019, May-not-Q4 peak, defensive promos) are each supported by an explicit statistical test in Section 2, with effect sizes and  $p$ -values; Sections 3–4 provide the remaining findings and the forecasting roadmap.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed?

Answer: [\[Yes\]](#)

Justification: Section 4 discusses data (missing SKU/region, 0.3% reconciliation error, 80%+ nulls in some promo columns), analytical (causal-agnostic correlations, right-censoring under stockout), and competition-specific limitations (548-day horizon exceeds one annual cycle).

### 3. Theory assumptions and proofs

Answer: [\[NA\]](#)

Justification: The paper reports empirical EDA; no theorem is stated.

### 4. Experimental result reproducibility

Answer: [\[Yes\]](#)

Justification: Appendix D lists the exact Python and LaTeX commands; every number in the paper is persisted in `metrics_v4.json` and regenerated deterministically by `eda_v4.py`.

### 5. Open access to data and code

Answer: [\[Yes\]](#)

Justification: Code (`eda_v4.py`, `eda_v4.ipynb`, `Makefile`), all figures (vector PDF plus PNG), and `metrics_v4.json` are provided in `results/v4/`. The dataset is supplied by the Datathon 2026 organisers per the contest rules.

### 6. Experimental setting/details

Answer: [\[Yes\]](#)

Justification: Section 4 specifies the rolling-origin CV folds, the full exogenous feature list, and the candidate models for Part 3; Appendix A documents table-level row counts and missingness.

### 7. Experiment statistical significance

Answer: [\[Yes\]](#)

Justification: Welch  $t$ -test (Section 2.1), Mann-Whitney  $U$  (Section 3.2), Spearman correlation (Sections 3.3–3.4) are reported with test statistics,  $p$ -values, and sample sizes. Figure 2A uses 95% confidence intervals; Cohen’s  $d$  is reported for the main regime effect.

### 8. Experiments compute resources

Answer: [\[Yes\]](#)

Justification: The EDA pipeline runs in under 10 seconds on a commodity laptop (CPU only; no GPU;  $\sim 1$  GB RAM peak) per Appendix D.

### 9. Code of ethics

Answer: [\[Yes\]](#)

Justification: No human-subjects research, no crowdsourcing, no PII handling; only anonymised customer IDs provided by the organisers are used.

### 10. Broader impacts

Answer: [\[Yes\]](#)

Justification: Prescriptive recommendations (predictive promo triggers, VIP retention) are framed as internal operational improvements; no deployment of generative or decision-making systems that would affect third parties.

### 11. Safeguards

Answer: [\[NA\]](#)

Justification: No pretrained models or scraped datasets are released.

**12. Licenses for existing assets**

Answer: [\[Yes\]](#)

Justification: The dataset is used under the Datathon 2026 competition terms (Reference [1]); all Python libraries cited in Appendix D are permissive open source (BSD/MIT/PSF).

**13. New assets**

Answer: [\[Yes\]](#)

Justification: The produced assets (the Python pipeline, figure PDFs, metrics JSON, notebook) are documented in `results/v4/summary.md` and shipped in the same folder as this report.

**14. Crowdsourcing and research with human subjects**

Answer: [\[NA\]](#)

Justification: The study does not involve crowdsourcing or human subjects.

**15. Institutional review board (IRB) approvals or equivalent**

Answer: [\[NA\]](#)

Justification: No human-subjects research is involved; IRB review is not applicable.

**16. Declaration of LLM usage**

Answer: [\[NA\]](#)

Justification: LLMs were not used as a component of the EDA methodology. Any LLM assistance was limited to writing and editing and does not affect the scientific methodology or the quantitative results.